

Aryan Bhatnagar

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EDUCATION

Stevens Institute of Technology, Hoboken, New Jersey

Expected May 2029

B.E. Mechanical Engineering, Co-op Program Track

GPA: 3.94 / 4.00

Awards: Dean's List, Pinnacle Scholar, Edwin A Stevens Scholar, Presidential Achievement Scholar

TECHNICAL SKILLS

CAD & Design: CATIA V5, SolidWorks (CSWA certified), Onshape, AutoCAD; 3D Modeling & 2D Drafting

Engineering Methods: GD&T, DFM, DFA, tolerance analysis, design for manufacturing and assembly, rapid prototyping, failure analysis design documentation

Analysis & Simulation: Heat transfer & thermal modeling, fluid mechanics simulation, FEA, MATLAB, Simulink

Controls & Programming: C/C++, Python, Arduino, ROS2, PID control

Prototyping & Fabrication: FDM 3D printing (PLA / ABS), hands-on assembly & bring-up, iterative prototyping

EXPERIENCE

EV / Aircraft Battery Thermal Management – Research Assistant

April 2025 - Present

Stevens Institute of Technology; Liquid Cold-Plate Design

- Designing a custom liquid cold plate in SolidWorks to hold an EV/aircraft battery below the 40 °C limit that guards against cell degradation and thermal runaway, dissipating a 500 W load through a 6-channel U-manifold sized for uniform flow.
- Built and validated a self-consistent ϵ -NTU wall-temperature solver (MATLAB), cross-checked against an LMTD energy balance and hand calculations, to benchmark three channel topologies; showed the U-manifold design meets the 500 W thermal duty at $\sim 6 - 9$ times lower parasitic pumping power than a serpentine baseline.
- Debugged and rebuilt the analysis pipeline by correcting specific-heat unit and extrapolation errors, per-channel bookkeeping, pump-efficiency handling, and the thermal-resistance reference, producing trustworthy figures of merit (h , NTU, ΔP , FOM) that drive coolant and geometry selection.
- Characterizing PAO-based nanofluid coolants for temperature-dependent properties (k , μ , cp , ρ) and iterating FDM-printed cold-plate geometries toward a bench-tested prototype to validate the model.

Stevens Robotics Club – Mechanical Design Lead

September 2025 - Present

VEXU Competition Team; Lead and Treasurer of 20-member team

- Lead mechanical design of a VEXU competition robot in SolidWorks, delivering custom 3D-printed sensor housings, modular battery mounts, and a rigid mechanical aligner for repeatable high-speed scoring; robot scored reliably throughout competition.
- Designed and integrated a high-pressure pneumatic actuation system and custom conveyor, optimizing cylinder placement and spatial tolerances for fast, reliable field deployment.
- Orchestrated comprehensive team financial operations and procurement logistics, strategically managing university budgets and purchase orders to fund iterative rapid prototyping cycles.

ENGINEERING PROJECTS

6-DOF Robotic Arm – Design Engineer

June 2026 - Present

- Designing a 6-DOF robotic arm from scratch in SolidWorks (PLA+ printed structure, ST3215 serial-bus servos), sizing joints, links, and load paths for stiffness and payload capacity.
- Developing inverse kinematics and PID trajectory control in ROS2 and MATLAB / Simulink so the arm follows commanded paths and coordinates, targeting autonomous pick-and-place of blocks.
- Integrating Arduino-based servo control and task / coordinate command handling for end-to-end actuation.

Autonomous Navigation Robot — Design Engineer

February 2026 - April 2026

- Designed and built a complete modular robot chassis in SolidWorks, packaging custom ultrasonic-sensor brackets and component mounts into a single fastener-based assembly.
- Applied DFM under a strict 4-hour print budget to eliminate non-essential geometry while preserving structural rigidity through snap-fit datums and fastener-located joints; completed the course in 40 s against an ~ 80 s field average ($\sim 2\times$ margin).
- Implemented waypoint path planning with PID heading correction driven by an overhead-camera ArUco-marker localization system streaming pose data to the ESP8266 over MQTT, enabling closed-loop navigation of the field.